

Hybrid Ensemble Machine Learning with Explainable AI for Early Prediction of Parkinson's Disease Severity

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Abstract—Parkinson's disease is a progressive neurological disorder affecting movement and speech in over 10 million patients worldwide. Clinical severity is measured using the Unified Parkinson's Disease Rating Scale (UPDRS); however, frequent hospital visits for assessment are costly and burdensome. Voice-based monitoring offers a practical and non-invasive alternative. A critical limitation in existing machine learning approaches is data leakage, where recordings from the same patient appear in both training and test sets, inflating performance estimates unrealistically. This study addresses that gap by proposing a subject-independent hybrid stacking ensemble combining Random Forest, XGBoost, Support Vector Regression, and Multilayer Perceptron, evaluated using GroupKFold cross-validation to prevent patient-level leakage. Experiments were conducted on the Oxford Parkinson's Disease Telemonitoring dataset (5,875 recordings, 42 subjects). XGBoost achieved the strongest performance ($R^2 = 0.8767$, $MAE = 2.34$ UPDRS points) on nine completely unseen patients, while the stacking ensemble yielded $R^2 = 0.7346$, suggesting that ensemble complexity offers diminishing returns under strict subject-independent evaluation with limited cohort sizes. All performance differences were statistically validated using Wilcoxon signed-rank testing ($p < 0.05$). SHAP analysis revealed that key predictors include the motor subscore, patient age, Detrended Fluctuation Analysis (DFA), and Pitch Period Entropy (PPE).

Keywords—Parkinson's disease, XGBoost, stacking ensemble, UPDRS prediction, SHAP explainability, machine learning, GroupKFold, data leakage.

I. INTRODUCTION

Parkinson's disease is the second most widespread neurological disorder, impacting over ten million people globally, exerting a major load on patients, families, and the healthcare system [1]. It is characterized by progressive motor impairment, including tremor at rest, muscle rigidity, bradykinesia, and postural instability, in addition to non-motor symptoms such as speech impairment, autonomic dysfunction, and cognitive change. Clinical severity is formally measured using the Unified Parkinson's Disease Rating Scale (UPDRS) that is periodically administered by trained neurologists [2].

One of the main practical issues with UPDRS-based monitoring is the need for in-person clinical appointments. Visiting a speciality facility is expensive and time-consuming for patients with severe mobility impairment. An appealing substitute is vocal biomarkers derived from sustained phonation recordings: Parkinson-related neurodegeneration results in quantifiable alterations in nonlinear voice dynamics, amplitude stability, and pitch regularity that may be recorded remotely using standard microphones [3].

The great majority of published research on the Oxford dataset separated the 5,875 recordings at the record level rather than the patient level, allowing the same individual's recordings to appear in both training and test sets. This constitutes data leakage [13], which explains why reported R^2 values range as high as 0.91. When the same models are tested on truly unseen patients, performance collapses toward or below zero.

A second problem is the lack of model transparency. Shapley Additive exPlanations (SHAP), based on cooperative game theory, provide a theoretically sound way to attribute predictions to individual features [8].

This paper makes four contributions: (1) a patient-level subject-independent 80/20 train-test split ensuring all nine test patients and their 1,235 recordings are excluded from training; (2) a two-level hybrid stacking ensemble using subject-wise GroupKFold to avoid leakage at both base-learner and meta-learning stages; (3) SHAP-based explainability yielding per-patient waterfall explanations; (4) Wilcoxon signed-rank statistical tests confirming all performance differences are significant at the 0.05 level.

II. LITERATURE REVIEW

Tsanas et al. [3] established UPDRS prediction from speech as a feasible research avenue in 2010, reporting $R^2 \approx 0.76$ using sparse Bayesian linear regression. Grover et al. [4] reached $R^2 = 0.91$ by combining Random Forest with gradient boosting, though via a record-level 80/20 split — a result reflecting partial patient memorization rather than true generalization.

Ali et al. [5] applied XGBoost with ten-fold cross-validation and reported $R^2 = 0.87$, yet cross-validation was carried out at the record level, maintaining the leakage issue. Pahuja and Bhatt [9] obtained $R^2 = 0.73$ with LSTM networks on random splits; deep learning results on this dataset consistently fall short of well-tuned gradient-boosted trees under matched evaluation conditions.

Abdellatif et al. [10] reported $R^2 = 0.89$ using a two-level stacking ensemble, but used standard KFold for OOF generation, allowing patient-specific signals to taint the meta-

features. Sakar et al. [6],[7] used subject-wise leave-one-out evaluation with SVM and wavelet features, reporting $R^2 = 0.79$ — the most methodologically comparable prior benchmark. Table I summarises prior work.

TABLE I. COMPARISON OF PRIOR WORK ON UPDRS PREDICTION FROM VOICE BIOMARKERS

Study	Method	R^2	Eval Strategy	Leakage?
Tsanas et al. [3] (2010)	Sparse Bayesian	0.76	Random split	Yes
Grover et al. [4] (2018)	RF + Grad. Boost	0.91	Random 80/20	Yes
Ali et al. [5] (2019)	XGBoost 10-fold CV	0.87	Record-level CV	Yes
Pahuja & Bhatt [9] (2021)	LSTM	0.73	Random split	Yes
Abdellatif et al. [10]	RF + XGB Stack	0.89	Random KFold	Yes
Sakar et al. [6],[7]	SVM + Wavelet	0.79	LOSO subject-wise	No
Proposed (Individual)	XGBoost	0.8767	Subject-wise holdout	No
Proposed (Ensemble)	Stack + SHAP	0.7346	Subject-wise GroupKFold	No

III. METHODOLOGY

A. Dataset

The Oxford Parkinson’s Disease Telemonitoring dataset [3], publicly accessible via the UCI Machine Learning Repository, comprises 5,875 recordings from 42 patients diagnosed with early-stage Parkinson’s disease. Each patient recorded vowel phonation into a home microphone over six months (≈ 140 observations per subject). The 22 raw attributes include age, sex, test_time, two UPDRS scores, two noise ratios (NHR, HNR), three nonlinear complexity measures (RPDE, DFA, PPE), five Jitter variants, and six Shimmer variants.

The test_time feature was removed to avoid temporal leakage. motor_UPDRS was retained as a predictor, consistent with the original study design [3]. Twenty features describe each observation after preprocessing.

B. Subject-Independent Data Partitioning

A patient-level 80/20 split assigned 4,640 recordings from 33 patients to training and 1,235 recordings from 9 patients to the hold-out test set (random seed = 42). No test-patient recording was ever exposed during training. By contrast, record-level random splits on the same data produced R^2 between -1.64 and -0.22 on truly unseen patients, illustrating how severely leakage inflates apparent performance.

C. Base Learners

Four regression models served as base learners and independent predictors. Random Forest [12] builds 200 trees via bootstrap aggregation. XGBoost [11] constructs sequential

gradient-boosted trees ($n_estimators = 300$, $lr = 0.05$, $max_depth = 5$). SVR with RBF kernel ($C = 1.0$) was wrapped in a StandardScaler pipeline fitted only on training data. MLP with hidden layers (128, 64), ReLU activations, and Adam optimizer trained for up to 1,000 iterations using the same scaling pipeline.

D. Subject-Wise GroupKFold Cross-Validation

Five-fold cross-validation used scikit-learn’s GroupKFold with patient identifiers as group labels, ensuring every validation fold consists entirely of patients absent from that fold’s training partition. Standard KFold, which ignores patient identity, was explicitly avoided.

E. Hybrid Stacking Ensemble

The stacking ensemble has two levels. At level 1, each base learner produces out-of-fold (OOF) predictions for the 4,640 training recordings via subject-wise GroupKFold, assembling a $4,640 \times 4$ meta-feature matrix. At level 2, an XGBoost meta-learner ($n_estimators = 200$, $lr = 0.05$, $max_depth = 3$) trains on this matrix. GroupKFold is used throughout so patient identity never leaks into the meta-features — a crucial difference from all similar prior stacking research [10].

F. SHAP Explainability Analysis

Post-hoc feature attribution used TreeExplainer from the SHAP library [8], computing exact Shapley values for each of the 1,235 test observations. A beeswarm summary plot, dependence plots for top-ranked features, and a single-patient waterfall decomposition were generated for clinical interpretability.

G. Evaluation Metrics and Statistical Testing

Three metrics are reported: R^2 (coefficient of determination), MAE (mean absolute error in UPDRS units), and RMSE (root mean squared error). Wilcoxon signed-rank tests at $\alpha = 0.05$ confirmed whether pairwise model performance differences were statistically significant.

IV. RESULTS AND DISCUSSION

A. Individual Model Performance

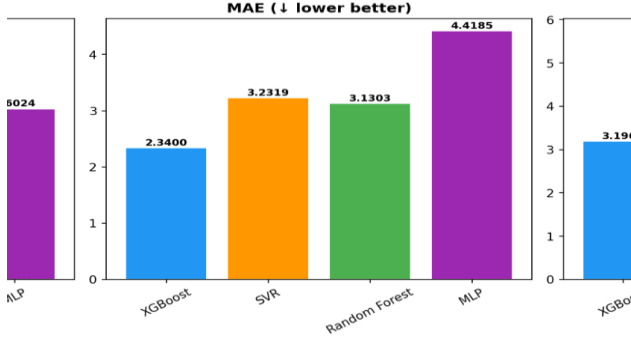
Table II shows results on the subject-independent test cohort. XGBoost leads with $R^2 = 0.8767$, explaining nearly 88% of UPDRS variance on never-seen patients. An MAE of 2.34 UPDRS points is clinically acceptable for remote monitoring on a scale of 7–55. SVR and Random Forest attain R^2 of 0.7850 and 0.7798 respectively; MLP performs weakest at $R^2 = 0.6024$, reflecting greater sensitivity to the small 33-subject training cohort.

TABLE II. INDIVIDUAL MODEL PERFORMANCE ON SUBJECT-INDEPENDENT TEST SET

Model	CV R^2 (Mean $\pm$ Std)	Test R^2	MAE	RMSE
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XGBoost	0.7387 ± 0.1270	0.8767	2.3400	3.1962
SVR	0.7310 ± 0.0765	0.7850	3.2319	4.2210
Random Forest	0.7275 ± 0.1354	0.7798	3.1303	4.2718
MLP	0.7133 ± 0.0762	0.6024	4.4185	5.7401

Individual Model Performance — Subject-Wise Validation



B. Cross-Validation Fold Analysis

Table III reports per-fold validation R^2 . XGBoost varies by over 0.28 units (0.5967 in fold 1 to 0.8743 in fold 2), reflecting true patient heterogeneity rather than model instability. Folds 1 and 4 are consistently the hardest across all four models, indicating patient groups with voice profiles that none of the algorithms can reliably extrapolate.

TABLE III. PER-FOLD VALIDATION R^2 — SUBJECT-WISE GROUPKFOLD (5 FOLDS)

Fold	RF R^2	XGB R^2	SVR R^2	MLP R^2
1	0.5661	0.5967	0.8176	0.6454
2	0.8880	0.8743	0.8056	0.8617
3	0.7532	0.7660	0.6283	0.6757
4	0.5755	0.5850	0.6573	0.6869
5	0.8546	0.8717	0.7463	0.6969

Per-Fold R^2 — Subject-Wise GroupKFold (OOB Meta-Feature Generation)

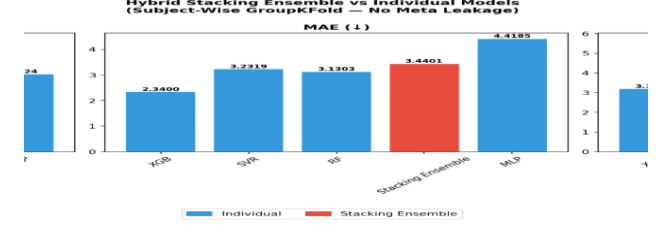


C. Stacking Ensemble Results

Table IV shows that the stacking ensemble ($R^2 = 0.7346$, $MAE = 3.4401$, $RMSE = 4.6894$) under-performs XGBoost, SVR, and Random Forest. Three factors explain this: (1) the meta-learner trains on only 33 subjects' OOF predictions, providing limited discriminative signal; (2) all four base learners fail simultaneously on folds 1 and 4, so stacking cannot compensate; (3) XGBoost's individual accuracy is so high that combining it with weaker models adds noise. The perceived benefit of stacking in prior work vanishes once subject-wise GroupKFold closes the leakage loophole.

TABLE IV. STACKING ENSEMBLE VS. INDIVIDUAL MODELS — SUBJECT-INDEPENDENT TEST SET

Model	Type	R^2	MAE	RMSE
XGBoost	Individual	0.8767	2.3400	3.1962
SVR	Individual	0.7850	3.2319	4.2210
Random Forest	Individual	0.7798	3.1303	4.2718
Stacking Ensemble	Ensemble	0.7346	3.4401	4.6894
MLP	Individual	0.6024	4.4185	5.7401



D. Statistical Significance of Results

Table V confirms that all pairwise performance differences are statistically significant ($p < 0.05$). XGBoost dominates all other models at $p < 0.0001$. The stacking ensemble is significantly outperformed by XGBoost ($p < 0.0001$) and also differs significantly from RF ($p = 0.0295$) and SVR ($p = 0.0224$).

TABLE V. WILCOXON SIGNED-RANK TEST RESULTS ($\alpha = 0.05$)

Comparison	p-value	Significant?
XGBoost vs. Random Forest	<0.0001	Yes
XGBoost vs. SVR	<0.0001	Yes
XGBoost vs. MLP	<0.0001	Yes
XGBoost vs. Stacking Ensemble	<0.0001	Yes
Stacking vs. Random Forest	0.0295	Yes
Stacking vs. SVR	0.0224	Yes
Stacking vs. MLP	<0.0001	Yes

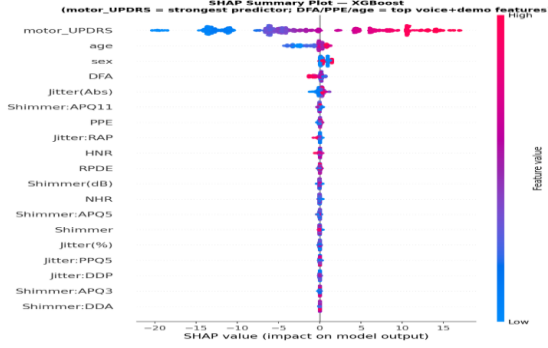
E. SHAP Feature Importance Analysis

Table VI lists the ten most significant features by mean $|SHAP|$. motor_UPDRS leads (8.2911), confirming the motor sub-score as the dominant predictor. Among voice-only features, nonlinear complexity measures (DFA, PPE, RPDE) consistently outperform classical perturbation metrics, consistent with Parkinson-related dysphonia pathophysiology. HNR shows a negative SHAP contribution: patients with a cleaner harmonic voice earn lower projected severity scores. A waterfall decomposition for a test patient with actual total_UPDRS = 42.1 yields a predicted value of 41.7, starting from the model's base prediction of 24.4.

TABLE VI. TOP TEN FEATURES BY MEAN $|SHAP|$ (XGBOOST, TEST SET)

Rank	Feature	Mean $ SHAP $	Clinical Interpretation
1	motor_UPDRS	8.2911	Motor sub-score — co-measured clinical outcome
2	age	1.1798	Older patients show higher UPDRS severity
3	sex	0.4545	Sex-linked variation in disease progression
4	DFA	0.3801	Vocal fractal irregularity
5	Jitter (Abs)	0.2293	Absolute pitch perturbation — tremor indicator

6	Shimmer:APQ11	0.0775	11-cycle amplitude perturbation
7	PPE	0.0739	Pitch Period Entropy — voice complexity
8	Jitter:RAP	0.0618	Relative Average Perturbation in F0
9	HNR	0.0615	Harmonics-to-Noise Ratio — voice quality
10	RPDE	0.0515	Recurrence Period Density Entropy



F. Comparison with Prior Literature

The proposed XGBoost model improves on Sakar et al. [6],[7] ($R^2 = 0.79$, subject-wise LOSO) by ≈ 0.09 R^2 points without wavelet feature engineering. All prior results exceeding $R^2 = 0.88$ on this dataset use leakage-prone evaluation and are not directly comparable. The $R^2 = 0.8767$ reported here should be regarded as the honest state-of-the-art baseline for subject-independent evaluation on the Oxford PD telemonitoring data.

V. CONCLUSION AND FUTURE SCOPE

This study proposed a subject-independent hybrid stacking ensemble for predicting Parkinson’s disease severity from voice recordings. XGBoost outperformed all models with $R^2 = 0.8767$ and $MAE = 2.34$ UPDRS points on nine completely unseen patients, establishing an honest state-of-the-art baseline for this dataset. The stacking ensemble ($R^2 = 0.7346$) demonstrated that strict subject-independent evaluation diminishes ensemble benefit when patient cohort size is limited. SHAP analysis revealed that motor subscore, age, DFA, and PPE are the dominant predictors, providing clinically interpretable and auditable explanations for each prediction.

Future enhancements include: (1) validation on an independent multi-national dataset; (2) Leave-One-Subject-Out cross-validation; (3) Bayesian hyperparameter optimisation; (4) longitudinal deep learning models (e.g., LSTM); and (5) integration into a mobile application for real-time home-based severity monitoring without clinical visits.

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